

Assessment 3 brief



Assessment details

Unit Code Title	MATH1003 Quantitative Methods with Economics
Assessment Type	Case Study; Calculations
Due Date	Week 6, Sunday, 11:00 pm (NSW, AEST)
Weight	45%
Length / Duration	1000 - 1250 words
Individual / Group	Individual
Unit Learning Outcomes (ULOS)	ULO2, ULO3, ULO4

Task overview

You will recall that in Assessment 2 the Moti family received an offer to expand their restaurant business into Papua New Guinea, and want to know more about financial viability. As the Moti's business analyst, in this assessment, you will build on your analysis from Assessment 2, present your findings and make a final recommendation to the Moti's. Your previous analysis found that the low season presents the largest risk to the financial viability of the business. The aim of this assessment to determine if the low season risk can be mitigated through simple strategies to increase income and reduce costs. As the business analyst, you will assess whether low season risk can be mitigated by:

- increasing revenue using menu promotions, thereby affecting demand; and
- implementing a seasonal menu to reduce the cost of ingredients.

Upon completing this analysis, you will present your findings and recommendation to the Moti's. You will provide a specific recommendation as to whether the restaurant is financially viable and would be a suitable investment for the family.

For this assessment you apply (i) quantitative skills from Modules 3, 4 and 5 and (ii) ethical considerations from Module 6. Your goal is to provide further well-informed business information communicate these recommendations to the family. You will complete the following three activities:

- **Part A: Increasing Revenue (Supply and Demand)**
 - Determine equilibrium price and quantity values for baseline conditions and factoring in a price reduction for a specified tourism season
 - **Part B: Reducing Restaurant Expenses (Simple and Laspeyres Indices)**
 - Determine Simple Indices and Laspeyres Indices for local produce used across three tourism seasons
 - **Part C: Recommendation to Moti's**
 - Based on your analyses from Part A and Part B AND your analyses from Assessment 2, provide a final Recommendation Report to the Moti's regarding the business proposal
 - Use GenAI to act as the Moti's and ask questions about your analyses
-

Rationale

In this assessment, you will continue in your role as a data analyst for the Moti Family Restaurants to support their decision regarding the investment in the new restaurant. You will analyse further operational data using skills from Modules 3-6, before presenting your recommendation to the Moti family. This task is designed to enhance your proficiency in assessing changes in supply and demand, calculating indices, and working with correlative data, whilst still emphasising the ethical use of Generative AI tools. You will build further upon your Assessment 2 submission to demonstrate your ability to interpret data, formulate data-driven recommendations, and assess the financial implications of business decisions, as well as presenting your recommendation and explanation to the Moti family.

Resources

To complete this assessment, you will need the following resources.

- Module 3
 - Module 4
 - Module 5
 - Module 6
-

Task Instructions

Part A. Revenue growth strategy (14 marks)

The analysis in Assessment 2 showed that the low reason presents the biggest financial risk due to lower revenue and increased variability. As the Moti's business analyst, you decide to model the effect of a low season special whereby customers who order a main meal get a complementary entrée or dessert could potentially increase revenue.

The restaurant operator's supply and demand functions for the low season are represented by the following equations:

- Demand equation: $p = 142 - 1.07q$
- Supply equation: $p = 0.0331q^2 + 0.06q + 12$
- q is the average number of customers
- p is the average amount of money spent by customers (for a two-course meal i.e. main meal + an entrée or dessert) in PGK

You model the proposed change by adjusting the **demand equation** to account for a 50 PGK price reduction applied to the cost of a two-course dinner. Answer the following:

- Calculate the **equilibrium p** , **equilibrium q** , and **daily revenue** outcome for baseline conditions (using the above equations). **Explain and show all calculations and workings.**
- Calculate the **equilibrium p** , **equilibrium q** , **daily revenue**, and the **price the customer pays** following implementation of the price reduction (for which you will modify the demand equation). **Show all calculations and workings.**
- What is the difference in the price paid by the customer? Express the change in **daily revenue** as a percentage change. **Show all calculations and workings.**
- Plot the supply, original demand, and revised demand equations on a single graph. Include a title, clearly labelled axes (including units) which are appropriately scaled, & label each function.
- What does this analysis show? (~200 words)
 - What is the overall effect of the promotion on revenue?
 - What is the impact on customers in terms of the price paid?
 - Why doesn't all of the price reduction get passed on to customers?
 - Discuss positive and negative effects of the change.

In completing this part of the task, ensure your responses address these three aspects:

- **Calculations:**
 - Calculate equilibrium points and revenue outcomes.
 - Present your calculations and results in a clear and organised manner.
- **Visual Representation:**
 - Plot Supply and Demand functions on the same set of axes
 - Ensure plots are clear, accurately labelled, and effectively convey data.
- **Data Interpretation:**
 - Interpret your findings by identifying the difference in revenue with the adjusted demand.
 - Explain the positive and negative aspects of the proposed change.

Part B. Expense reduction strategy (13 marks)

A major expense for any restaurant operator is the ongoing cost of purchasing ingredients for menu items, which can fluctuate considerably due to a variety of factors, including food inflation. Food inflation refers to changes in food prices and can be influenced by many factors including (but not restricted to):

- **Shifting Supply and Demand Dynamics:** Seasonal variations in agricultural activity and productivity can directly affect the price of harvested food items.
- **Supply Chain Problems:** Disruptions to transportation can increase the cost of food, with transportation companies often passing on the costs of any disruptions to deliveries.

As the business analyst, you investigate the seasonal cost of key ingredients, including three staple main meal ingredients (sweet potato, taro & yam) and three key dessert ingredients (banana, sago, coconut). Table 2 represents the average price (per kg) and approximate quantities of these ingredients purchased by the restaurant operator during low, shoulder and high seasons.

Table 2

Average cost (per kilogram; PGK) and quantity purchased (kg) of common restaurant ingredients across seasons

	SEASON	HIGH	HIGH	SHOULDER	SHOULDER	LOW	LOW
Course	Staple ingredients	Price/kg (PGK)	Quantity (kg)	Price/kg (PGK)	Quantity (kg)	Price/kg (PGK)	Quantity (kg)
Mains	Sweet Potato	1.37	1556	1.73	1275	3.31	766
Mains	Taro	2.24	732	2.39	590	2.99	328
Mains	Yam	4.21	1025	4.51	822	5.27	482
Desserts	Banana	1.76	466	2.04	380	6.99	239
Desserts	Sago	2.55	1191	2.76	946	3.02	568
Desserts	Coconut	3.12	532	3.28	424	3.46	255

(i) Calculate the Laspeyres Price Index (to three decimal places) for Low and Shoulder Seasons, using the High Season as the base. **Show all workings.**

(ii) Calculate the simple price indices for each individual food item for Low and Shoulder Seasons, using the High Season as the base. **Show all workings.**

(iii) From this, determine which food item(s) are most responsible for the difference in the Laspeyres Price Index values calculated in part (i) for both seasons.

(iv) Select **one ingredient** from the Low Season to partially substitute for another similar ingredient (see Table 2). Choose a percentage (multiple of 10, i.e. 20%, 70%) to decrease your selected ingredient. Calculate the **number of kilograms** of this percentage decrease and match this reduction with an **identical increase** in the

substitution ingredient. For example, if you reduce an ingredient by 10%, which amounts to 25 kilograms, you must add 25 kilograms to the substitution ingredient. **Show all workings.**

(v) Calculate the **total original cost** of all Low Season ingredients and then the **revised total cost** after you make your substitution. Calculate the difference in total cost and express this as a percentage change. **Show all workings.**

(vi) What e. what would or could change in terms of items offered to potential customers?

(vii) Your analyses shows certain operational changes can affect revenue and ingredient costs. Using your **new revenue** calculations from Part A(ii), **new ingredient cost** calculations from Part B(v), and assuming **additional weekly costs of 31590 PGK** (this includes additional ingredients, staff wages, utility costs, insurance etc), calculate the projected **weekly profit**. **Show all workings.**

(viii) How does this compare to the **average** and **maximum profit** values obtained from the analysis in Assessment 2?

In completing this part of the task, ensure your responses address these three aspects:

• **Calculations:**

- Calculate Simple and Laspeyres Indices.
- Calculate costings for ingredients at season level where appropriate.
- Present your calculations and results in a clear and organised manner.

• **Trend Identification:**

- Analyse the indices of each food item (using High Season as base)
- Identify and describe trends apparent (relative to High Season as base)
- Interpret the impact specific food items have on and contribute towards the overall LPI values calculated and total seasonal cost.

• **Data Interpretation:**

- Provide evidence of written analysis of index values.
- Provide a summary of trends you have identified.
- State modifications to masses of ingredients purchased that could lead to a reduction in costs in the Low Season.

Part C. Final recommendation (18 marks)

Recommendation Report (12 marks, 900 words)

You have now analysed all of the data and are ready to provide a recommendation regarding the business proposal. Using your analyses from both **Assessment 2 AND Assessment 3**, write a **Recommendation Report for the Moti family** summarising your analyses, findings, and providing a **specific** recommendation. A template will be provided for this recommendation report. Your report must be structured as follows:

- Executive summary (200 words)
 - A brief summary of the body of the report
 - This section is usually written last, once the remainder of the report is complete.
- Summary of analyses/findings from Assessments 2 and 3 (400 words):
 - Weekly profit analysis (Assessment 2, Part A)
 - Financial modelling (Assessment 2, Part B)
 - Revenue growth strategy (Assessment 3, Part A)
 - Expense reduction strategy (Assessment 3. Part B)

- Do not present full calculations here – just present summary outcomes
- Include at least one graph
- Statement addressing ethical considerations and sustainability concerns arising from findings that may influence your recommendation (150 words):
 - Consider potential impacts on customers
 - Consider potential impacts on the environment
- Final recommendation to the Moti's on whether to invest or not to invest, with justification of why you are making this recommendation (150 words).

Tailor the language of this recommendation report to the Moti family: while they have a background running restaurants, they do not have a detailed understanding of the mathematics. **Do not give any background information about the family** in your report – they already know who they are!

Response to questions from Moti family (6 marks, 300 words)

You are likely to receive comments and questions from the Moti's regarding your analysis and recommendations. Simulate this by asking GenAI to role-play as the family and ask questions regarding your recommendation report.

To complete this section using GenAI to role-play as the Moti family:

1. Provide background information about the Case Study and Moti's to CoPilot (via your own prompt/s). Remember that you are writing to (and for) the Moti family. Include screenshots of all prompts and GenAI outputs.
2. Submit your **Recommendation Report only** to Microsoft CoPilot (**do not** submit your responses for Part A and B) and ask CoPilot (using your own prompt/s) to:
 - a. assume the role of the Moti family
 - b. ask five (5) questions relating to the analyses conducted and what the outcomes obtained mean.
3. Include **screenshots** of **all five (5) questions** provided.
4. Select **two (2)** of these questions and give a short answer to each one (i.e. maximum of 100 words per question).
5. Reflect on why you chose to respond to those two (2) questions (100 words).

Referencing

If you consult any external sources for this task, you must acknowledge them and use the Harvard referencing style for citations and referencing. For additional guidance, refer to the [SCU Library referencing guides](https://www.scu.edu.au/library/study/referencing-guides/) (<https://www.scu.edu.au/library/study/referencing-guides/>).

GenAI use

This assessment is classified as:

- **Level 2: AI use permitted for specific purposes.** You can use AI for specific tasks as identified by the UA.

 You may use GenAI to:

- Help structure your report.
- Help define key terminology.
- Brainstorm ideas for interpreting the data.
- Check grammar and spelling in your interpretation.
- Derive questions the Moti family could ask about your analyses.

 You may not use GenAI to:

- Analyse the data
 - Generate the plots or calculate the statistics.
 - Directly generate your interpretation.
 - Write your recommendation summary for the Moti family.
-

GenAI declaration

What is a GenAI declaration?

A GenAI declaration is a short statement included with your assessment that explains whether you used generative artificial intelligence tools when completing the task.

This assessment permits limited, preparatory use of GenAI. If you used any generative AI tools, you must declare how they were used in accordance with the permitted conditions outlined above.

The purpose of a GenAI declaration is to:

- promote academic integrity and transparency
- Clarify how digital tools supported your learning
- distinguish your own thinking from tool-assisted input
- ensure fair and consistent assessment

A declaration does not penalise you for using GenAI. Instead, it allows assessors to understand how you engaged with the task and whether your use aligns with the approved conditions.

How do I declare GenAI use?

You must choose **one** of the following options and include it at the end of your submission.

Option 1 - I used GenAI

If you knowingly used GenAI to complete any part of this assessment, you **must** include a brief statement that explains how you used GenAI in accordance with the use conditions outlined in this brief or stated by the Unit Assessor.

Your declaration **must** include:

- **The tools you used** - list the tools used (e.g. *Microsoft Copilot, ChatGPT 5.0, Grammarly Premium, etc.*).
- **How you used them** - a short note on what you asked the tool to do, and how you used the output in your work.

Example declaration

| For Assessment 3, I used CoPilot to clarify definitions and Grammarly to review grammar in my draft. All calculations, graphs, explanations, interpretations and recommendations are considerations are my own work. I also used CoPilot for Part C of this task when I asked it to role-play as the Moti family and provide me with a selection of questions to choose from, after inputting my summary report.

Your declaration must reflect the actual way you used the tool. Using AI in ways that were prohibited in this brief remains academic misconduct, even if declared.

Option 2 - I did not use GenAI

If you did not use GenAI to complete any part of this assessment, please include the following statement in your assessment submission:

"I confirm that I, [insert full name], have not knowingly or intentionally used generative AI to complete any part of this assessment."

Where do I include my declaration?

- Place your declaration at the **end of your main submission file** (e.g. the last page of your submission, immediately after your response for Part C).
 - Your GenAI declaration does **not** count towards the assessment word limit.
-

Submissions

What to submit

To complete this assignment, you must submit the following items:

- A single Word (.doc or .docx) file, via the submission portal provided for this task.

When to submit

SCU assessment deadlines are set to New South Wales (NSW) Time. Blackboard automatically converts this deadline to match your local time. Remember, the date and time displayed in the **assessment submission** portal is the due date adjusted for your location. You must submit your work **before** this displayed time.

How to Submit

To begin the submission process, navigate to **Assessment 3 - Submission** and select '**View Instructions**' to upload your file.

Assessment rubric

This rubric outlines the grading criteria against which your submission will be assessed, along with the standards expected for each grade level. For further information, refer to [Final Grades](https://www.scu.edu.au/current-students/student-administration/final-grades/) (<https://www.scu.edu.au/current-students/student-administration/final-grades/>).

👉 **Click to download a copy of the assessment rubric.**



Assessment 3 Marking Rubric.docx